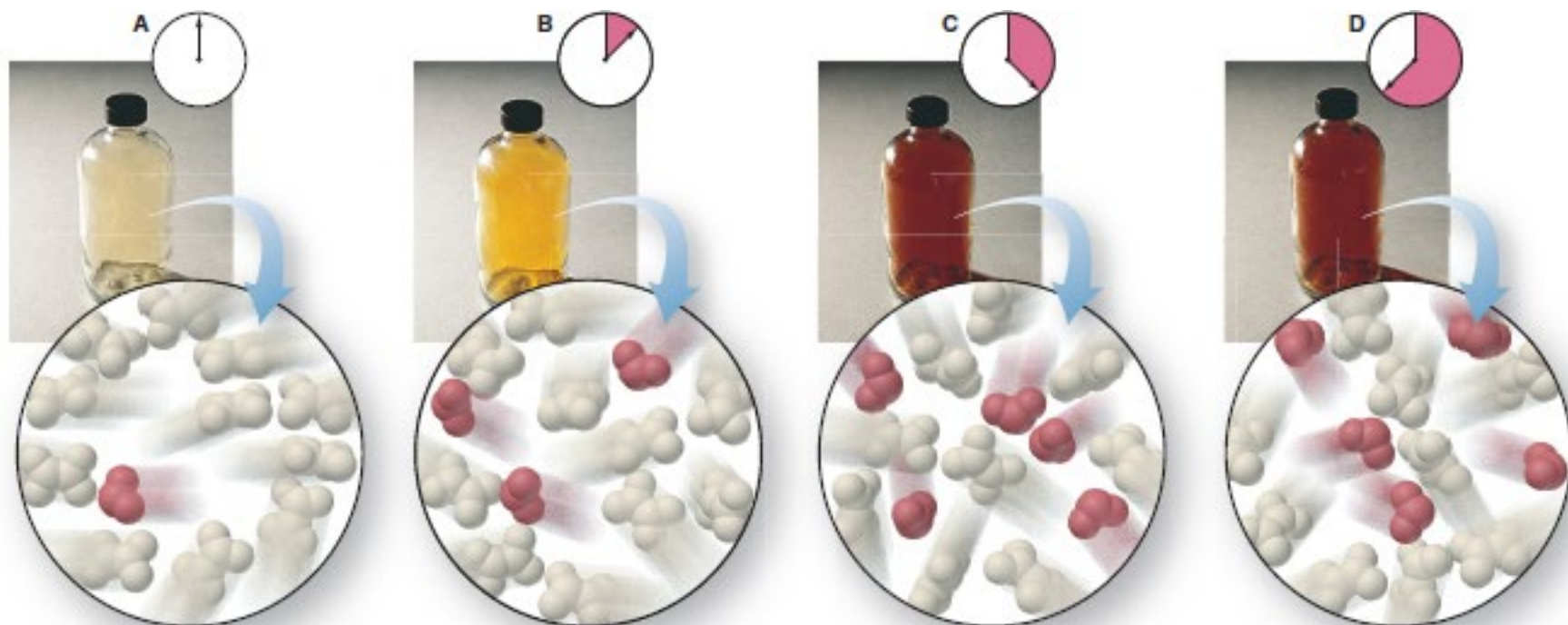


Chapter 17: Kinetics Content Lecture Pt. 2



KINETICS Part 2

Central focus:
The rates of chemical reactions
(How fast does the reaction proceed)

Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4: Integrated Rate Laws

17.5: Collision and Transition State Theory

17.6: Reaction Mechanisms

17.7: Catalysis

Determining Rate Law: Initial Rate Method

Determine initial rate – Concentration of reactants is known (product is negligible)

What is the rate law for the following reaction?



Experiments with different reactant initial concentrations and initial rates

Experiment	$[\text{NO}]_0$ (M)	$[\text{H}_2]_0$ (M)	Initial Rate (M/min)
1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
3	0.0100	0.0200	0.0120

We can only determine Rate Laws for an overall reaction via experiment

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	[NO] ₀ (M)	[H ₂] ₀ (M)	Initial Rate (M/min)
1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
3	0.0100	0.0200	0.0120

Write arbitrary rate in terms
Of reactants

$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^y$$

Between Exp 1 and 3: There is a change in concentration of H₂ which changes the rate – we can determine y

$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^y$$

$$0.006 = k [0.01]^x [0.01]^y$$

Divide top rate by bottom rate

$$0.0120 = k [0.01]^x [0.02]^y$$

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	$[\text{NO}]_0 (\text{M})$	$[\text{H}_2]_0 (\text{M})$	Initial Rate (M/min)
→ 1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
→ 3	0.0100	0.0200	0.0120

$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^y$$

Between Exp 1 and 3: There is a change in concentration of H_2 which changes the rate – we can determine y

$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^y$$

$$0.006 = k [0.01]^x [0.01]^y$$

$$0.0120 = k [0.01]^x [0.02]^y$$

$$0.5 = 0.5^y \quad y = 1$$

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	[NO] ₀ (M)	[H ₂] ₀ (M)	Initial Rate (M/min)
1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
3	0.0100	0.0200	0.0120

$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^1$$

Between Exp 1 and 2: There is a change in concentration of H₂ and NO which changes the rate - we know y=1 so now we can determine x

$$0.006 = k [0.01]^x [0.01]$$

$$0.144 = k [0.02]^x [0.03]$$

Divide top rate by bottom rate

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	[NO] ₀ (M)	[H ₂] ₀ (M)	Initial Rate (M/min)
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$$\text{Rate} = k [\text{NO}]^x [\text{H}_2]^1$$

Between Exp 1 and 2: There is a change in concentration of H₂ and NO which changes the rate - we know y=1 so now we can determine x

$$0.006 = k [0.01]^x [0.01]$$

$$0.144 = k [0.02]^x [0.03]$$

Divide top rate by bottom rate

$$0.04167 = 0.5^x \cdot 0.333333$$

$$0.1250 = 0.5^x$$

$$\ln(0.1250) = \ln(0.5^x) = x \cdot \ln(0.5)$$

$$-3 = x \cdot (-1) \quad \text{so } x = 3$$

$$\text{Rate} = k [\text{NO}]^3 [\text{H}_2]^1$$

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	$[\text{NO}]_0 \text{ (M)}$	$[\text{H}_2]_0 \text{ (M)}$	Initial Rate (M/min)
1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
3	0.0100	0.0200	0.0120

$$\text{Rate} = k [\text{NO}]^3 [\text{H}_2]$$

Can you determine k for this reaction?

Substitute in any Expt. Rate data into Rate Law

Determining Rate Law: Initial Rate Method

What is the rate law for the following reaction?



Experiment	$[\text{NO}]_0 (\text{M})$	$[\text{H}_2]_0 (\text{M})$	Initial Rate (M/min)
1	0.0100	0.0100	0.006
2	0.0200	0.0300	0.144
3	0.0100	0.0200	0.0120

$$\text{Rate} = k [\text{NO}]^3 [\text{H}_2]$$

Can you determine k for this reaction?

Note that the exponents don't match

The stoichiometric coefficients for this reaction

$$0.144 \text{ M min}^{-1} = k (0.02 \text{ M})^3 (0.03 \text{ M})$$

$$k = \frac{0.144 \text{ M min}^{-1}}{(0.02 \text{ M})^3 (0.03 \text{ M})} = 6.0 \times 10^5 \text{ M}^{-3} \text{ min}^{-1}$$

Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4: Integrated Rate Laws Would like to know rate at any time

If we know we have zero, first order, second order

17.5: Collision and Transition State Theory

17.6: Reaction Mechanisms

17.7: Catalysis

Integrated Rate Law: expresses concentration directly as a function of time
(want to know the full time evolution of the reaction)

Zero-Order Reaction



$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k [A]^0 = k$$

Units of $k = \text{M s}^{-1}$

Units of k must multiply to give M s^{-1} for Rate

$$k = -\frac{\Delta[A]}{\Delta t}$$

When we integrate this over time interval 0 to t

$$-\Delta[A] = k \Delta t$$

For an infinitesimal change

$$-dA = k dt$$

$$[A]_t - [A]_0 = -kt$$

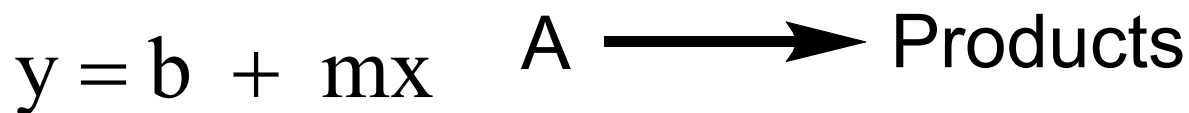
$$[A]_t = [A]_0 - kt$$

$$\int_{[A]_0}^{[A]_t} dA = -k \int_0^t dt$$

You **don't** have to know integration!

This simply shows where the time integrate rate law comes from

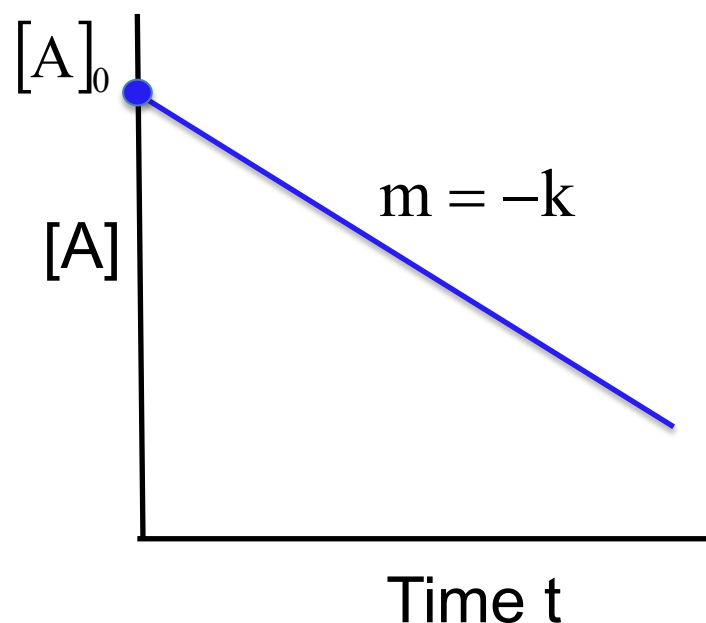
Zero-Order Reaction: determining k



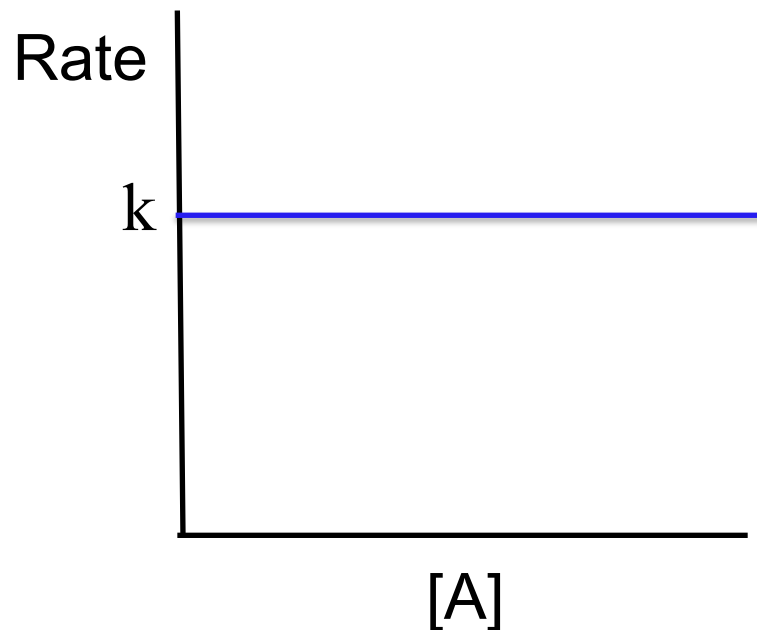
$$[A]_t = [A]_0 - kt$$

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k [A]^0 = k$$

Plot of $[A]$ as a function of time



Plot of rate as a function of $[A]$



First-Order Reaction



$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k [A]$$

Units of $k = \text{min}^{-1}; \text{s}^{-1}; \text{hr}^{-1}$ or
any inverse time unit

$$k [A] = -\frac{\Delta[A]}{\Delta t}$$

Units of k must multiply to give M s^{-1} or
 M t^{-1} for Rate ($t = \text{min}$ or $t = \text{hr}$ or $t = \text{yr}$)

$$k \Delta t = -\frac{\Delta[A]}{[A]}$$

For an infinitesimal change: When we integrate this
over time interval 0 to t

$$\frac{dA}{A} = -k dt$$

$$\int_{[A]_0}^{[A]_t} \frac{dA}{A} = -k \int_0^t dt$$

You **don't** have to know
Calculus derivations
For Chem120

$$\ln[A]_t - \ln[A]_0 = -kt$$

$$\ln[A]_t = -kt + \ln[A]_0$$

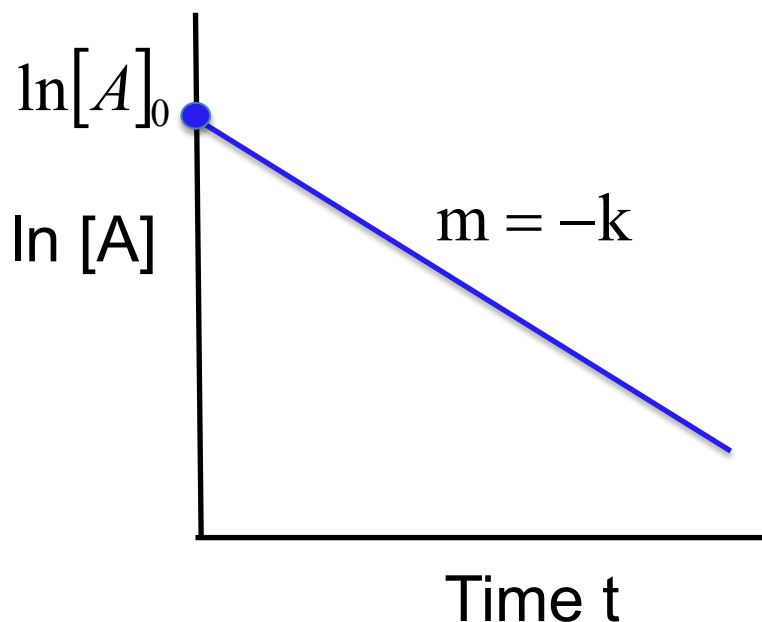
First-Order Reaction: determining k



$$\ln[A]_t = -kt + \ln[A]_0$$

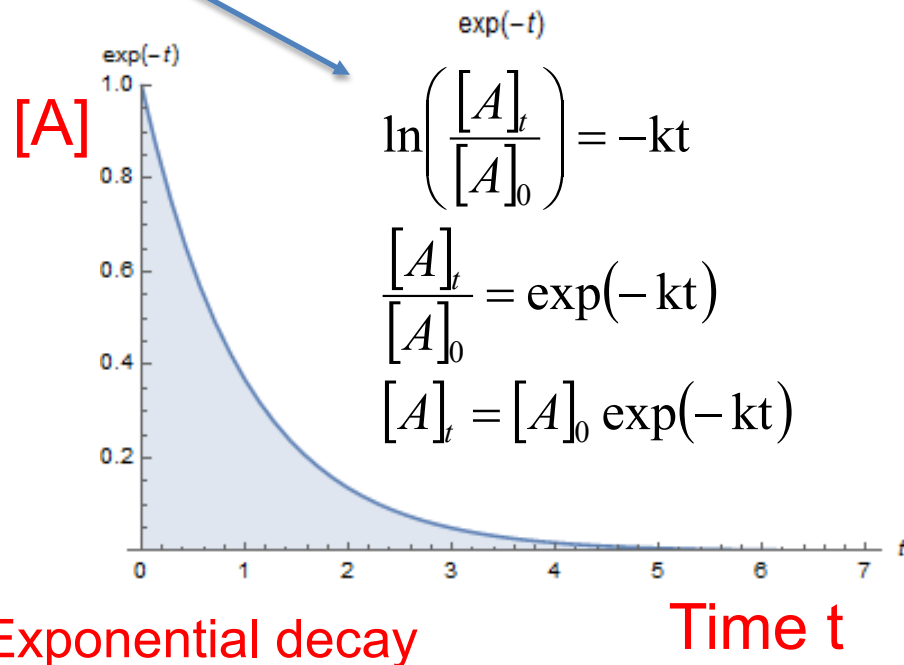
$$y = mx + b$$

Plot of $\ln[A]$ as a function of time



$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$$

Plot of $[A]$ versus time



$$\ln\left(\frac{[A]_t}{[A]_0}\right) = -kt$$
$$\frac{[A]_t}{[A]_0} = \exp(-kt)$$
$$[A]_t = [A]_0 \exp(-kt)$$

First-Order Reaction: determining k



$$\ln[A]_t = -kt + \ln[A]_0$$

$t_{1/2}$ is the time taken for one-half of the reactant to be consumed

At $t_{1/2}$ $[A]_{t_{1/2}} = \frac{1}{2} [A]_0$

$$\ln[A]_{t_{1/2}} = \ln\left(\frac{1}{2} [A]_0\right) = -kt_{1/2} + \ln[A]_0$$

$$+ kt_{1/2} = \ln[A]_0 - \ln\left(\frac{1}{2} [A]_0\right) = \ln\left(\frac{[A]_0}{\frac{1}{2} [A]_0}\right)$$

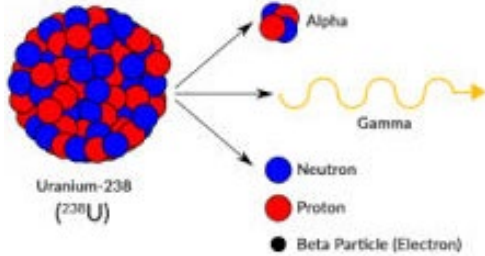
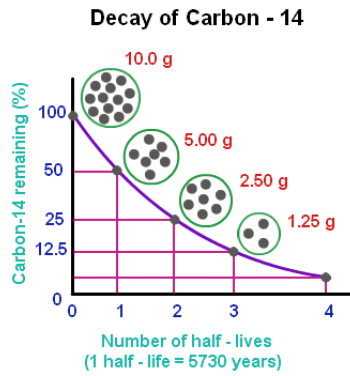
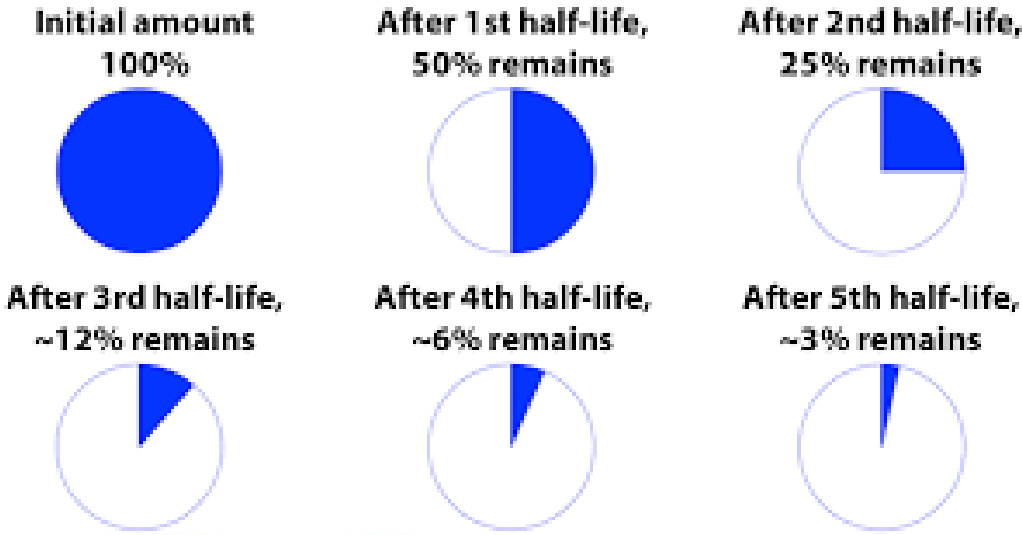
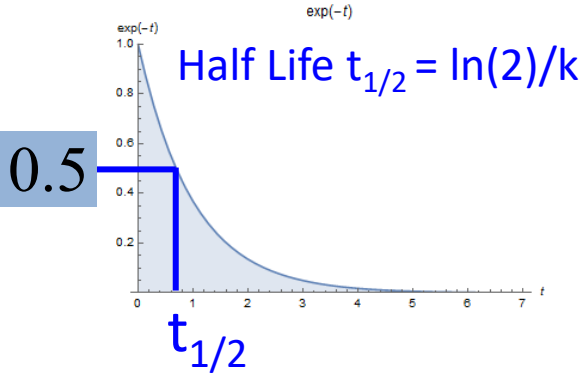
$$t_{1/2} = \frac{\ln(2)}{k} \approx \frac{0.6931}{k}$$

$$k = \frac{\ln(2)}{t_{1/2}}$$

First-Order Reaction: Half life of a reaction

First Order Reaction Half Life

Half life $t_{1/2}$ time for reaction to consume
Half of the initial concentration of product



Radioactive decay



Carbon dating
Radioactive dating with ^{14}C $t_{1/2} = \sim 5770$ years

Second-Order Reaction



$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k [A]^2$$

$$k [A]^2 = -\frac{\Delta[A]}{\Delta t}$$

$$\text{Units of } k = \text{M}^{-1} \text{ s}^{-1}$$

Units of k must multiply to give M t^{-1} for Rate

$$k \Delta t = -\frac{\Delta[A]}{[A]^2}$$

For an infinitesimal change:

$$\frac{dA}{A^2} = -k dt$$

$$\int_{[A]_0}^{[A]_t} \frac{dA}{A^2} = -k \int_0^t dt$$

$$\frac{1}{[A]_0} - \frac{1}{[A]_t} = -kt$$

When we integrate this over time interval 0 to t

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

You **don't** have to know
Calculus derivations
For Chem120

Second-Order Reaction: determining k

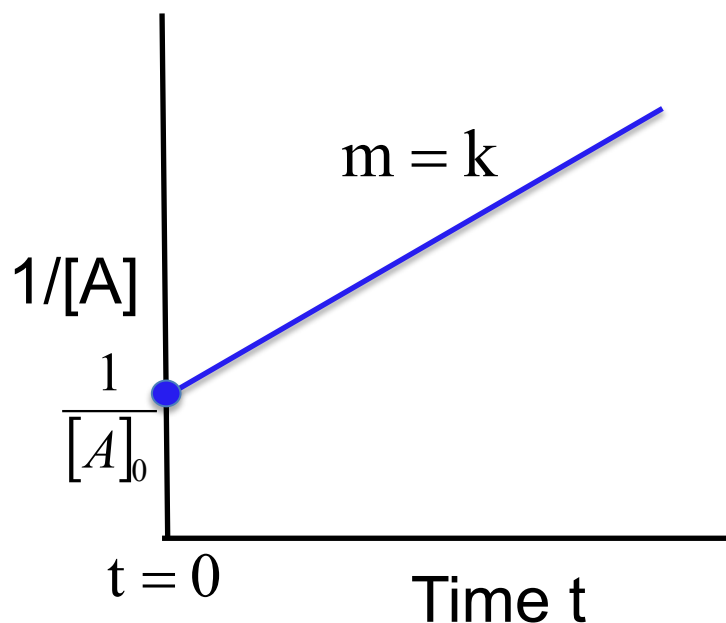
$$y = mx + b$$

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

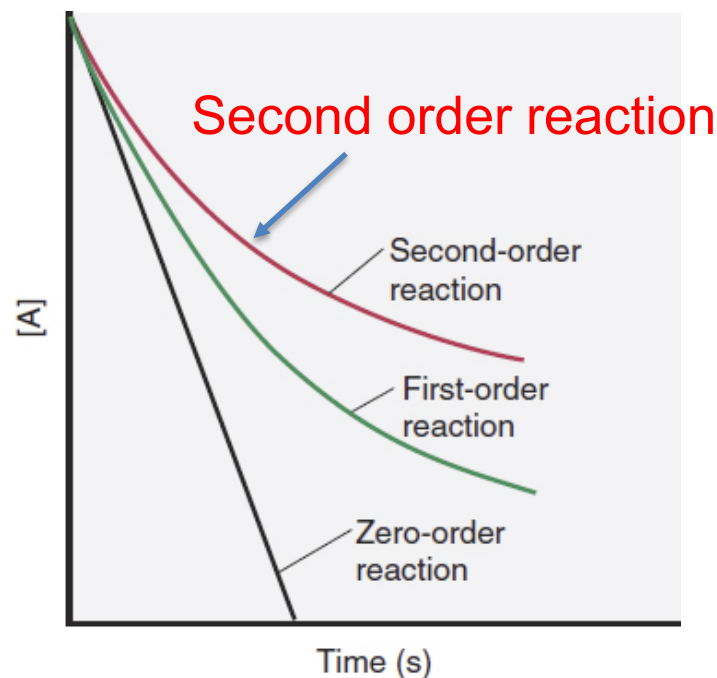


$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

Plot of $1/[A]$ as a function of time



Plot of $[A]$ versus time



Second-Order Reaction (Type II)



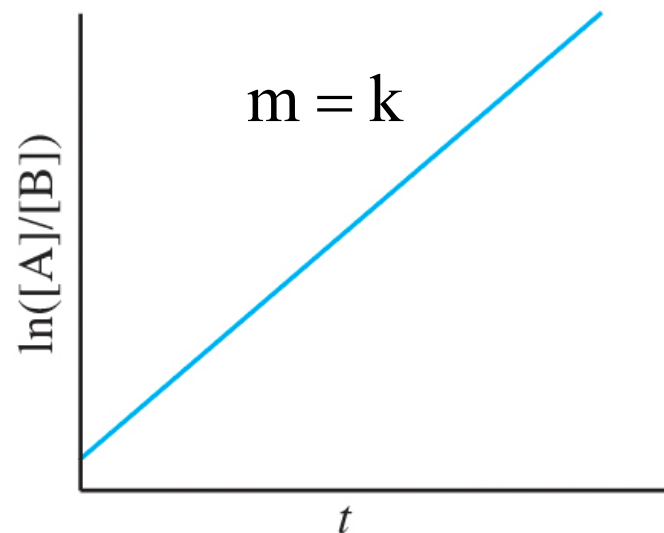
You are **not responsible** for 2nd Order
Type II for Chem 120

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} = k[A] \cdot [B]$$

$$\int \frac{dx}{(a-x)(b-x)} = \int k dt \Rightarrow \text{Nice integration!}$$



$$\frac{1}{[A]_0 - [B]_0} \ln \left(\frac{[B]_0 [A]}{[A]_0 [B]} \right) = kt \quad (\text{for } [A] \neq [B])$$



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Pseudo first-order reactions



$$\text{Rate} = k [A][B]$$

The above reaction is an example of a second order reaction
If there is an excessive amount of B, such that $[B] \gg [A]$

During the course of the reaction, the change in the concentration of B is negligible. We can modify the equation in such a case

$[B]$ can be considered to be constant

$$\text{Rate} = k [A][B]$$

$$\text{Rate} = k' [A]$$

Chapter 17: Kinetics

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Collision Theory: Effect of temperature/concentration

Ave. Kinetic Energy of molecules increases with Temp.

$$\overline{E_{KE}} = \frac{3}{2} \frac{RT}{N_A} = \frac{3}{2} k_b T$$

Number of collisions increases with concentration

$$P = \frac{1}{3} \left(\frac{N}{V} \right) m \overline{u^2}$$

Collision frequency can be calculated
From molecular kinetic theory



If each collision led to a reaction – then the rates of these reactions would be $\sim 10^6 \text{ Ms}^{-1}$

Rates of reactions of gases usually $\sim 10^{-4} \text{ Ms}^{-1}$ – so only a small fraction of gas molecule collisions result in a reaction

Collision Theory: Effect of temperature

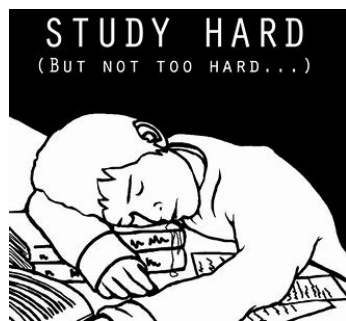
Effect of temperature on the rate of a reaction

In order to form products:

- Bonds need to be broken and formed (reorganized)
- This reorganization requires energy
- When molecules collide they must have a minimum kinetic energy (Activation Energy, E_a) to initiate the reaction

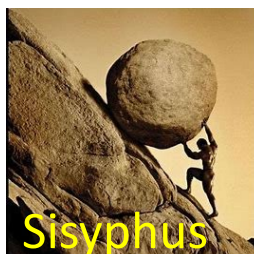


Svante Arrhenius



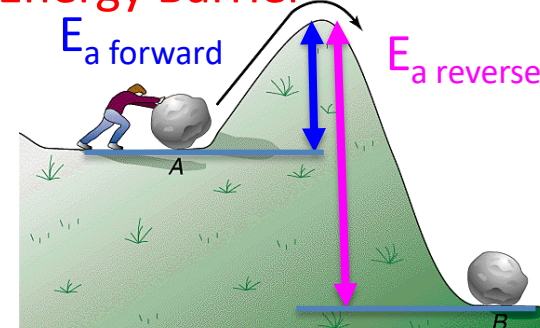
Chem 120

Concept of an activation Energy barrier is familiar to most of us!



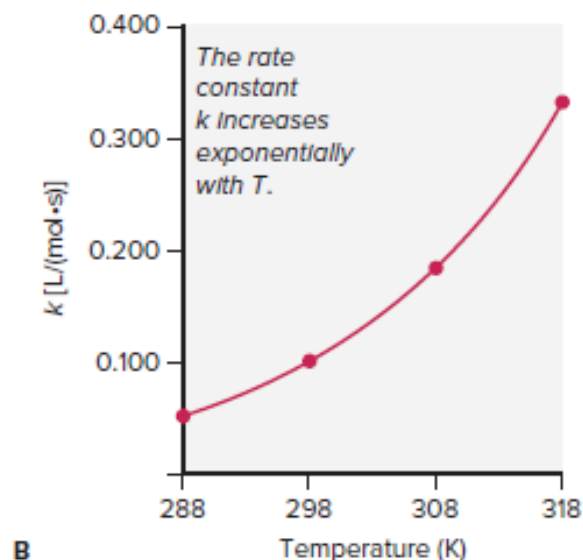
Sisyphus

Activation Energy Barrier



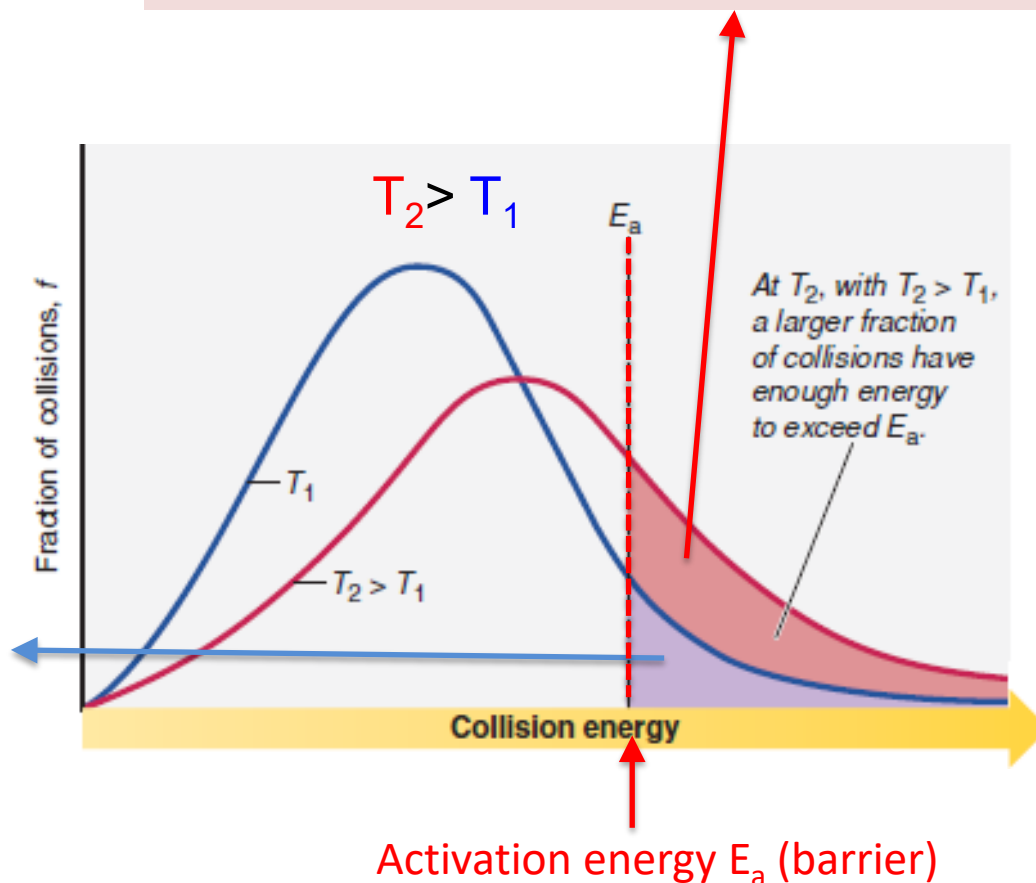
Original Univ. of Florida Chemistry

Effect of temperature on collision frequency and energy



At lower temperature T_1 , a lower fraction of molecules have the activation energy required

At higher temperature T_2 , a higher fraction of molecules have the activation energy required



Arrhenius Equation

Rate of the reaction depends upon:

- Collision Frequency
- Orientation of the molecules
- Fraction of molecules with $KE > E_a$

$$k = Ae^{-\frac{E_a}{RT}} \quad \text{Arrhenius Equation}$$

Rate constant k ...
larger k faster rate

A : pre-exponential factor

R : Universal gas constant

E_a : Activation energy

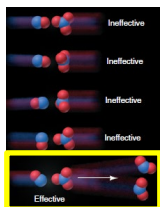


Svante Arrhenius
Nobel (Chemistry) 1903

A : pre-exponential factor

Arrhenius constant

Related to collision frequency
& molecular orientation



Recall from Chp. 8

$$\text{Collision } \nu = \frac{u_{Ave}}{mfp} = 7.1 \times 10^9 \text{ collision s}^{-1}$$

 N_2 (g) at $P=1$ bar $T=293.15$ K

Arrhenius Equation

Taming the [Arrhenius] Equation.

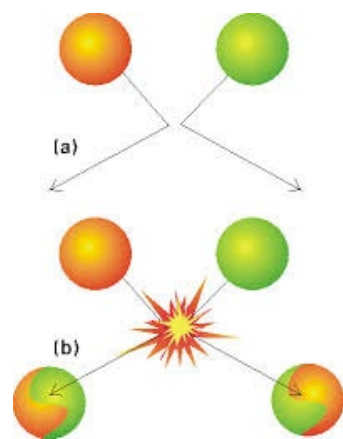
$$k = Ae^{-\frac{E_a}{RT}}$$



*Acknowledge Dr. Hans Christian von Baeyer for the Little Prince analogy of taming

Big picture...rate constant temperature dependence...Negative exponential function

Molecular picture...

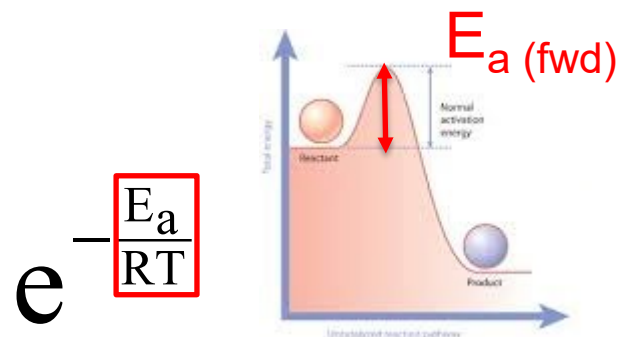


Molecular collisions

Frequency & orientation (A)

Kinetic Energy of molecules

$$E_{KE} \propto RT$$



$$e^{-\frac{E_a}{RT}}$$

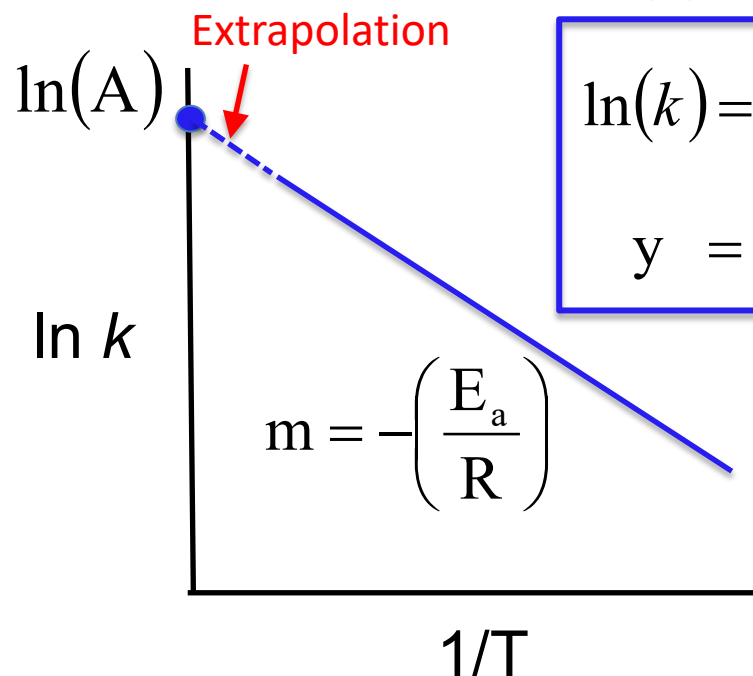
Temperature controls
Fraction of molecules
With enough KE to get
Over the barrier

Determining E_a experimentally

Plot of $\ln k$
as a function of $1/T$

$$k = Ae^{-\frac{E_a}{RT}}$$

$$\ln(k) = \ln\left(Ae^{-\frac{E_a}{RT}}\right) = \ln(A) + \ln\left(e^{-\frac{E_a}{RT}}\right) = \ln(A) - \frac{E_a}{RT}$$



$$\ln(k) = -\left(\frac{E_a}{R}\right)\frac{1}{T} + \ln(A)$$
$$y = m \cdot x + b$$

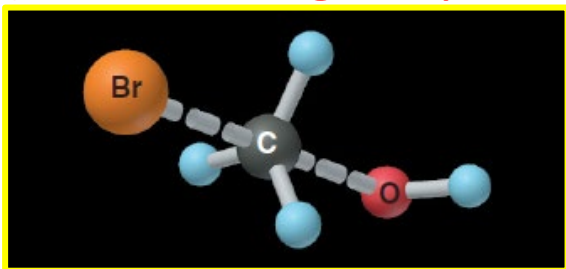
$$R = 8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$$

Change temperature T_1 to T_2

$$\ln\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{R}\right)\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

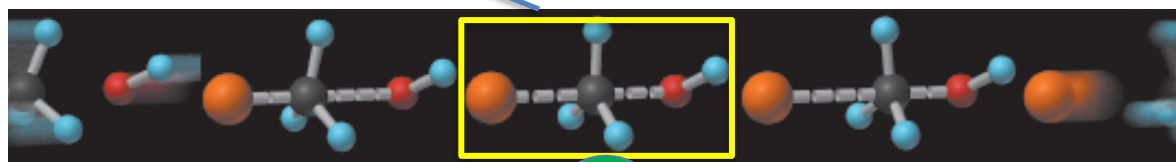
Transition State Theory

Transition state (TS)
(unstable higher potential energy)



Note bond forming C---O
Note bond breaking Br---C

However, it is possible that
The reaction does not go forward
And it can revert back to reactants



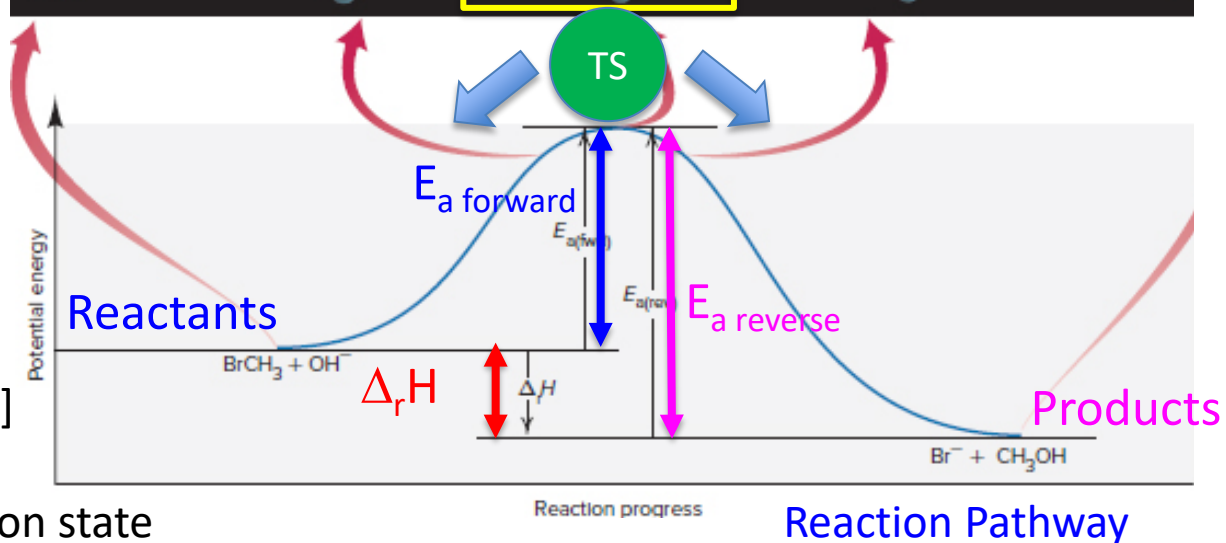
$$k = \frac{k_b T}{h} e^{-\frac{\Delta G^\ddagger}{RT}}$$

Eyring Equation

k_b = Boltzmann constant [J/K]

h = Planck constant [J s]

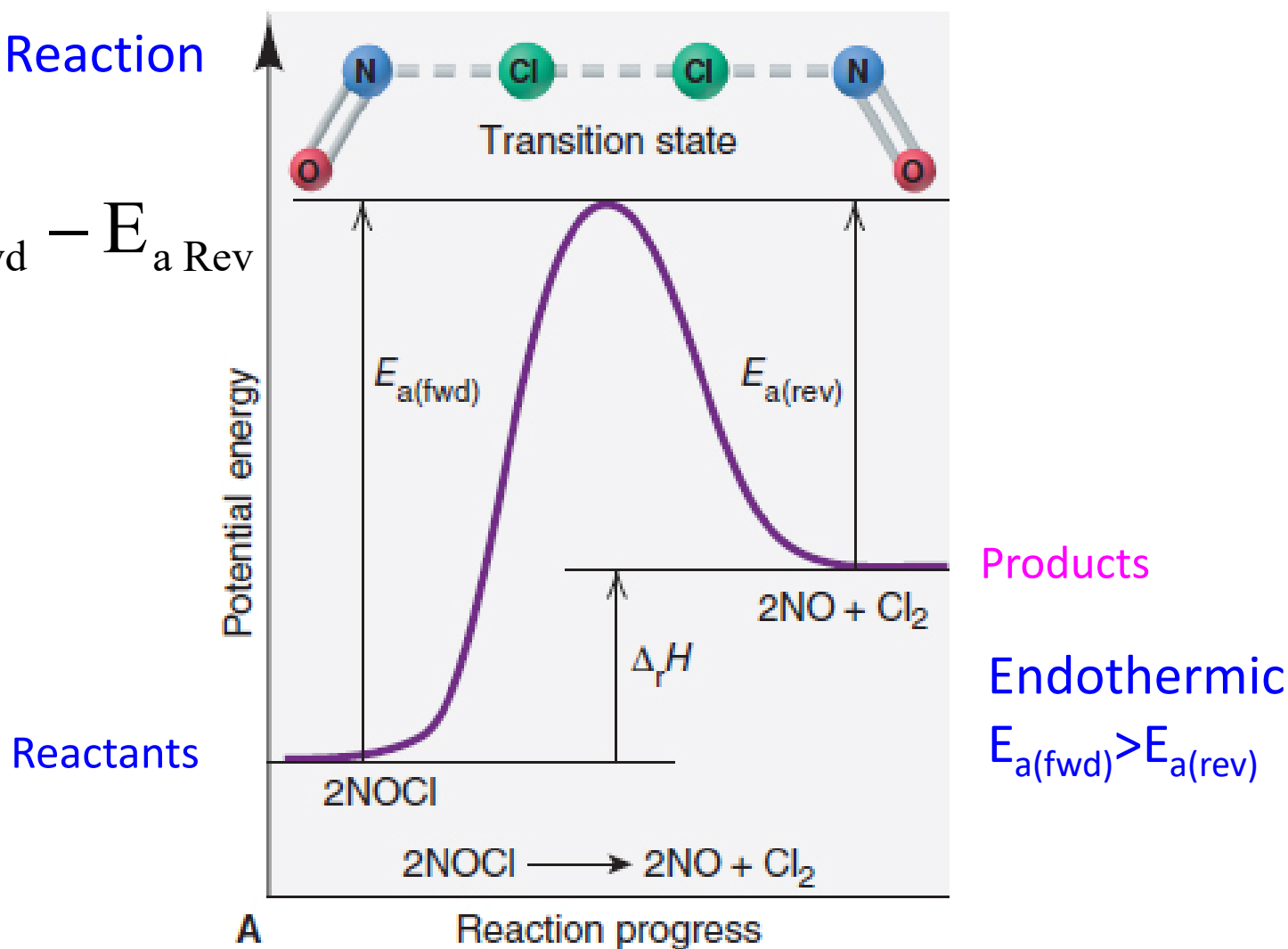
$\Delta G^\ddagger$ = Free energy of transition state



Transition State Theory

Endothermic Reaction

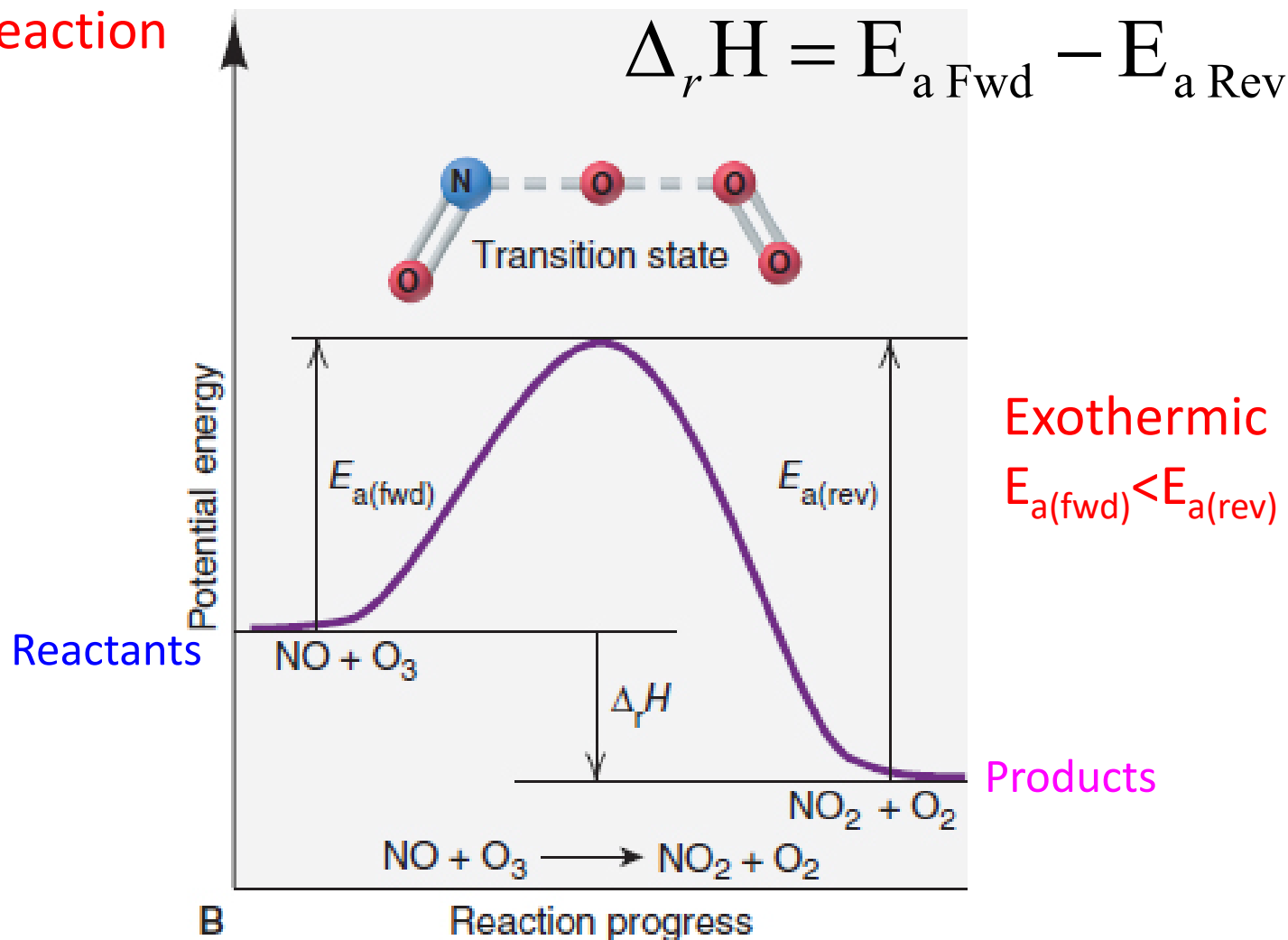
$$\Delta_r H = E_{a \text{ Fwd}} - E_{a \text{ Rev}}$$



$$\Delta_r H = \Delta H_{\text{Products}} - \Delta H_{\text{Reactants}} > 0$$

Transition State Theory

Exothermic Reaction



$$\Delta_r H = \Delta H_{\text{Products}} - \Delta H_{\text{Reactants}} < 0$$